

Intelligent Cooperative Sensing for Connected and Autonomous Vehicles: An Improved Decision Transformer Approach

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Abstract—Effective sensing capabilities are crucial for the safe and reliable operation of Connected and autonomous vehicles (CAVs). While traditional approaches focus on enhancing onboard sensors, the integration of road sensor networks (RSNs) into the CAV ecosystem presents a promising solution to improve sensing performance, but also introduces significant challenges, including heterogeneous sensing requirements, inconsistencies in multisource sensor data, and efficient resource utilization. To address these challenges, this article proposes a novel cooperative sensing framework that leverages multisource and multilevel sensing information from RSNs to optimize CAV sensing performance in resource-constrained scenarios. We develop an improved decision transformer (DT)-based approach that dynamically adapts to diverse driving conditions and efficiently fuses sensor data at various abstraction levels. To tackle the issue of long-delayed rewards, we introduce a reshaped reward function and a bi-level optimization framework that enables effective propagation of rewards along decision sequences. An advanced gradient approximation technique is employed to efficiently solve the optimization problem. Extensive simulations demonstrate the superior performance of our improved DT approach compared to state-of-the-art reinforcement learning (RL) methods in terms of sensing accuracy, coverage, and data efficiency under various traffic conditions.

Index Terms—Connected and autonomous vehicles (CAVs), cooperative sensing, decision transformer (DT), vehicular networks.

I. INTRODUCTION

THE ADVENT of Connected and autonomous vehicles (CAVs) is revolutionizing the automotive industry,

Received 18 August 2024; revised 10 December 2024; accepted 6 January 2025. Date of current version 23 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62231020; in part by the Innovation Capability Support Program of Shaanxi under Grant 2024RS-CXTD-01; and in part by the Technology Innovation Leading Program of Shaanxi under Grant 2023KXJ-116. (Corresponding author: Yuchuan Fu.)

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Digital Object Identifier 10.1109/IIOT.2025.3530857

offering the potential for increased safety, efficiency, and convenience [1]. Ensuring the reliable and safe operation of CAVs heavily relies on effective sensing capabilities [2], [3], [4]. Traditional approaches to improve vehicle sensing focus on enhancing the onboard sensors of individual vehicles, such as increasing the number and quality of sensors like cameras, LiDAR, and radar [5], [6]. While these efforts have led to significant advancements, the limited installation space and size constraints of onboard sensors pose a significant challenge in further improving the perception range and accuracy of individual vehicles solely through the enhancement of onboard sensing capabilities [7].

The emergence of road sensor networks (RSNs) provides a promising solution to mitigate the limitations of onboard sensing [8], [9]. By deploying sensors on roadside infrastructure, such as traffic lights, street lamps, and buildings, RSNs can provide vehicles comprehensive and enhanced sensing capabilities [4], [5]. These strategically deployed sensors can gather valuable information about the surrounding environment, including the presence of obstacles, road conditions, and the behavior of other road users [3]. Sharing this information with CAVs enables a collaborative sensing approach that effectively eliminates potential unseen area and improves the overall perception of the driving environment, ultimately enhancing safety and efficiency [10]. However, the integration of RSNs into the autonomous driving ecosystem presents several significant challenges.

These challenges can be categorized into three main areas. First, the heterogeneous nature of CAVs demands varying levels of sensing accuracy and range based on driving scenarios, autonomy levels, and priorities (comfort, efficiency, or safety) [2], [7]. This diversity complicates the design of effective sensing assistance toward CAVs. Second, multisource sensors in RSNs are susceptible to external factors like weather and lighting, potentially leading to inconsistencies or inaccuracies in transmitted information [11], [12]. This variability can result in incorrect driving decisions if not properly managed [9]. Third, the efficient utilization of computing and communication resources is crucial, given the diverse capabilities of vehicles and RSUs [13], [14]. Adaptive strategies for processing and transmitting sensor data at various abstraction levels (raw, feature, or result) are necessary to balance data quality and resource utilization [7]. Addressing these challenges requires an intelligent decision-making process that

can effectively integrate and utilize multisource and multilevel sensing information while considering the resource constraints and dynamic nature of the driving environment.

The decision transformer (DT), a method integrating reinforcement learning (RL) and contextual conditional sequence modeling [15], presents a promising solution to the challenges in RSNs. DT's multihead attention mechanism allows the model to attend to information from different representation subspaces, enabling a comprehensive and adaptive decision-making process that considers multiple factors in driving decisions [8]. However, one significant limitation of DT is its performance degradation when dealing with long-delayed rewards. In the context of RSNs, the impact of sensing decisions on the overall driving performance may not be immediately apparent, leading to delayed feedback and sparse rewards. This poses a challenge for traditional DT architectures, as they struggle to effectively propagate the delayed rewards back to the relevant decision steps [16], resulting in suboptimal learning and decision making.

Despite the promising potential of cooperative sensing in CAVs, existing approaches face challenges in accommodating heterogeneous sensing requirements, effectively integrating multisource sensor data, and optimally allocating limited resources in dynamic environments. In an attempt to address the above challenges, we propose a novel DT-based approach that optimizes cooperative sensing performance in RSNs by effectively leveraging multisource and multilevel sensing information while considering the resource constraints of the driving environment. Our approach introduces a reshaped reward function that enables the DT to handle long-delayed rewards effectively. By dynamically adapting to the current driving scenario and efficiently fusing information from multiple sensors at different levels of abstraction, our DT-based approach empowers vehicles to make informed decisions and optimize their sensing performance in real-time. Our main contributions are summarized as follows.

- 1) We propose a novel cooperative sensing framework that intelligently integrates multilevel and multisource sensing information to optimize onboard sensing performance in resource-constrained driving environments. Our approach dynamically adapts to the current physical driving conditions, available sensing modalities, and diverse driving demands of vehicles, enabling a more comprehensive and adaptive decision-making process.
- 2) We develop a concrete DT algorithm based on our proposed framework, which efficiently utilizes available resources and fuses information from multiple sensors at different levels of abstraction. This algorithm enables vehicles to make informed decisions and optimize their sensing performance in real-time, considering the dynamic nature of the driving scenario.
- 3) We introduce a customized reward function that addresses the performance degradation problem of DT when dealing with long-delayed rewards. By reshaping the reward distribution across the decision timeline, our approach enables DT to effectively capture the long-term impact of sensing decisions and optimize its performance in environments with sparse and delayed feedback.

- 4) We extensively validate the performance of our DT-based approach through both simulations and real-world experiments. The results demonstrate the superior sensing performance of our approach compared to state-of-the-art baselines, highlighting its effectiveness in optimizing sensing decisions in dynamic driving scenarios. We provide a comprehensive analysis of the experimental results, offering insights into the behavior and adaptability of our approach under various driving conditions.

The remainder of this article is organized as follows. We review related literature in Section II. Section III presents the system model and problem formulation. Section IV introduces our proposed DT-enabled cooperative sensing strategy. In Section V, we present our detailed design toward improved DT. Section VI discusses the simulation results, demonstrating the effectiveness of our approach in comparison to existing methods. Finally, Section VII concludes this article.

II. RELATED WORK

This section provides a comprehensive review of the existing literature relevant to this article's focus. It covers three main areas: 1) cooperative sensing; 2) decision making and optimization in resource-constrained environments; and 3) DT.

A. Cooperative Sensing

Cooperative sensing has emerged as a promising approach to enhance the perception capabilities of CAVs in vehicular networks [17], [18]. Ye et al. [17] proposed an accuracy-aware and resource-efficient cooperative sensing and computing scheme that enables fine-grained sensing data selection, transmission, fusion, and processing. Brambilla et al. [18] presented a distributed processing technique to augment the performance of global navigation satellite systems (GNSS) by leveraging vehicle-to-everything (V2X) communication systems.

Despite the promising results, cooperative sensing still faces several challenges and limitations. Xu et al. [19] identified the need for quantitative metrics to measure the quality of vehicular cyber-physical systems (VCPSs) and proposed a new metric called age of view (AoV). Wang et al. [20] focused on the efficient upload and dissemination of sensory data, particularly in resource-constrained environments, and proposed a deduplication-oriented mutual-assisted cooperative video upload method. Liu et al. [10] addressed the challenge of efficient interaction and management of heterogeneous crowd agents in mobile crowd sensing systems.

While cooperative sensing has shown great potential in enhancing the perception capabilities of CAVs, there are still significant challenges and limitations that need to be addressed, including resource constraints, heterogeneous sensing requirements, efficient data transmission and fusion, and the management of diverse sensing entities.

B. Decision Making and Optimization in Resource-Constrained Environments

Decision making and optimization in resource-constrained environments are critical aspects of cooperative sensing systems, particularly in vehicular networks. Recent studies

have focused on various approaches to address these challenges, such as multiscale RL for optimizing deep neural network models in resource-constrained Artificial Intelligence of Things (AIoT) [21], unified frameworks for joint sensing and communication in mobile crowd sensing systems [14], and vehicular edge federated learning solutions that leverage connected vehicles' onboard resources [22]. Wu et al. [21] proposed a multiscale RL approach for optimizing deep neural network models in resource-constrained AIoT. Their method leverages the hierarchical structure of neural networks to enable efficient optimization under limited resources.

Other works have investigated resource-constrained replication strategies for hierarchical and heterogeneous tasks in large-scale cloud computing systems [23], federated domain adaptation frameworks that extend domain adaptation with the constraints of federated learning while enhancing computation and communication efficiency [24], and deep RL-based computation offloading in resource-constrained edge computing systems [16]. Lu et al. [25] developed a federated domain adaptation framework that addresses the challenges of non-IID data in mobile edge computing (MEC) while enhancing computation and communication efficiency. Additionally, researchers have proposed joint edge-AI training and inference strategies that capture generalization and personalization for efficient inference across resource-constrained clients and DNN splitting frameworks for running DNN-based classification tasks on concurrent battery-based sensors [26].

These studies highlight the importance of efficient decision making and optimization techniques in resource-constrained environments to ensure optimal performance and resource utilization. However, challenges remain in addressing the diverse requirements and constraints of cooperative sensing systems, particularly in dynamic and heterogeneous vehicular networks.

C. Decision Transformer

DT has emerged as a powerful tool for enhancing decision-making processes across various domains. By casting offline RL as a sequence modeling problem, DTs derive the desired expected action from the goal and reward-to-go of future rewards, past states, and actions [15]. This approach has been successfully applied in several fields, including computer vision, gaming, healthcare, and recommendation systems.

In computer vision, the vision transformer (ViT) model has achieved comparable outcomes to top CNN-based models for image classification by leveraging large amounts of data and incorporating the transformer module [27]. In the gaming domain, multigame decision-making approaches using transformer-based models have been developed to optimize strategies and learn patterns applicable to various gaming environments [28]. Zhang et al. [29] proposed a continuous-time DT for healthcare applications, demonstrating its effectiveness in capturing the temporal dynamics and making informed decisions in real-time. The success of DTs in diverse fields suggests their potential for addressing the challenges in intelligent cooperative sensing for CAVs. Furthermore, DTs have been explored in recommendation systems to improve user

TABLE I
MAIN NOTATIONS USED IN OUR PROPOSAL

Notation	Description
$\mathcal{N}$	Set of connected and autonomous vehicles
N	Number of CAVs
$\mathcal{M}$	Set of roadside units
M	Number of RSUs
$\mathcal{S}$	Set of sensors
$\mathcal{A}$	Set of target sensing areas
$\mathcal{C}$	Set of sensing content
$\mathcal{D}_r$	Raw sensory data
$\mathcal{D}_f$	Feature-level sensory data
$\mathcal{D}_l$	Result-level sensory data
$\mathcal{R}_m$	Computing resources on RSU m
$\mathcal{R}_v$	Computing resources on CAV v
$\mathcal{B}$	Available communication bandwidth
P^D	Sensor deployment locations
$C(A_i^T, s_j)$	Sensing probability of s_j for A_i^T
$GC(A_i^T, \mathcal{S})$	Global sensing coverage for A_i^T
Decision variables	
$x_{c,v,m}$	Content c transmission: RSU m to CAV v
$y_{c,m}^k$	Content c processing on RSU m , resource k
$z_{c,v}^k$	Content c processing on CAV v , resource k
n_i^j	Sensors deployed at location p_i^D

retention by analyzing and predicting user preferences and behaviors [11].

These diverse applications demonstrate the versatility and potential of DTs as powerful AI models capable of adapting to different challenges and improving outcomes in various environments. However, further research is needed to fully understand the limitations and potential drawbacks of DTs in each domain and to develop techniques for effectively integrating them into existing decision-making processes.

III. SYSTEM MODEL AND PROBLEM FORMULATION

This section presents a detailed description of the system under consideration, including the key components, their characteristics, and interactions. The sensing content model and the RSNs model are formally defined. The section culminates in a precise mathematical formulation of the multiobjective optimization problem that this article aims to solve. The main notations are summarized in Table I.

A. System Overview

The system involves a set of CAVs denoted as $\mathcal{N} = \{1, \dots, n, \dots, N\}$. Each CAV is equipped with an advanced cellular vehicle-to-everything (C-V2X) integrated onboard unit (OBU), enabling robust vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications [22]. These vehicles are also outfitted with onboard sensors and computing units, allowing them to perceive their surroundings and make autonomous decisions [29]. The CAVs have heterogeneous driving requirements, which can be classified into comfort ($\mathcal{R}_c$), travel efficiency ($\mathcal{R}_e$), and safety ($\mathcal{R}_s$) [2]. Comfort-oriented CAVs prioritize passenger comfort, aiming for a smooth and pleasant riding experience. Travel efficiency-focused CAVs optimize travel time and fuel efficiency, prioritizing shorter routes and adapting driving behavior to minimize energy consumption. Safety-oriented CAVs prioritize collision avoidance and adherence to traffic rules,

requiring more accurate and more reliable, even redundant, sensing information to ensure the safety of passengers and other road users. Please note that in this article, users who prioritize efficiency and comfort do not necessarily ignore vehicle safety. On the contrary, our work is based on the personalized needs of users while ensuring vehicle safety [30].

A network of roadside units (RSUs) deployed along the roadways as part of the MEC infrastructure, represented as $\mathcal{M} = \{1, \dots, m, \dots, M\}$. Each MEC node, associated with a 6G base station, is stationary and equipped with substantial computational power and storage capacity [16]. These nodes offer edge computing, edge caching, and V2I communication services to CAVs within their coverage area, defined by a circular region with a radius of radius_m [31]. This setup enables efficient communication among vehicles and between vehicles and infrastructure, facilitating sensing assistance.

B. Sensing Content Model

The multisource sensors deployed on the roadside infrastructure generate diverse sensing content, which can be classified into three levels: 1) raw sensory data ($\mathcal{D}_r$); 2) feature-level sensory data ($\mathcal{D}_f$); and 3) result-level sensory data ($\mathcal{D}_l$) [7]. Each piece of sensing content $c \in \mathcal{C}$ is associated with a specific sensing requirement $o \in \mathcal{O}$ and can be represented as a tuple $c = (d_r, d_f, d_l)$ where $d_r \in \mathcal{D}_r$, $d_f \in \mathcal{D}_f$, and $d_l \in \mathcal{D}_l$.

Raw sensory data (d_r) refers to the unprocessed data directly captured by the sensors, such as camera images, radar signals, or lidar point clouds. Feature-level sensory data (d_f) is obtained by processing the raw data to extract relevant features or patterns that can be used for perception and decision making. This may involve techniques, such as object detection, segmentation, or tracking. Result-level sensory data (d_l) represents high-level insights or decisions derived from further analysis of the feature-level data, such as object classification, trajectory prediction, or risk assessment.

The amount of data and the required processing power increase from raw sensory data to result-level sensory data [7]. The transmission and processing of sensing content are subject to the available computing resources on the RSUs and available communication resources can be represented as $\mathcal{R}_m = \{r_m^1, r_m^2, \dots, r_m^K\}$ where r_m^k denotes the available computing resource of type k (e.g., CPU, GPU, and memory) on RSU m , and K is the total number of resource types [3]. Similarly, the available computing resources on each CAV $v \in \mathcal{V}$ can be represented as $\mathcal{R}_v = \{r_v^1, r_v^2, \dots, r_v^K\}$. The available communication bandwidth between a CAV v and an RSU m can be denoted as $b_{v,m} \in \mathcal{B}$ where $b_{v,m}$ represents the available bandwidth for the communication link between CAV v and RSU m .

C. Road Sensor Networks Model

The RSNs consist of multisource sensors $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ deployed at various locations $\mathcal{P}^D = \{p_1^D, p_2^D, \dots, p_M^D\}$ along the roadside infrastructure. Each sensor $s_j \in \mathcal{S}$ is characterized by its sensing probability function, which depends on factors, such as deployment

height, sensing distance, sensing pan, sensing tilt, and sensing obstacles.

The sensing probability of a single sensor s_j deployed at location p_i^D for covering a target sensing area A_i^T is given by

$$C(A_i^T, s_j) = \theta_{iD}^{\text{Dis}}(x_j) \cdot \xi_{iD}^{\text{Hor}}(x_j) \cdot \tau_{iD}^{\text{Ver}}(x_j) \cdot \varepsilon_{iD}^{\text{Obs}}(x_j) \quad (1)$$

where

$$\theta_{iD}^{\text{Dis}}(x_j) = 1 - \frac{1}{1 + e^{-\alpha(|x_{i,j}^{\text{Dis}} - x_{i,j}^D| - \beta)}} \quad (2a)$$

$$\xi_{iD}^{\text{Hor}}(x_j) = \frac{1}{1 + e^{-\iota(x_{i,j}^{\text{Hor}} + \eta)}} - \frac{1}{1 + e^{-\iota(x_{i,j} - \eta)}} \quad (2b)$$

$$\tau_{iD}^{\text{Ver}}(x_j) = \frac{1}{1 + e^{-\delta(x_{i,j}^{\text{Ver}} + \varrho)}} - \frac{1}{1 + e^{-\delta(x_{i,j} - \varrho)}} \quad (2c)$$

$$\varepsilon_{iD}^{\text{Obs}}(x_j) = \begin{cases} \frac{G}{|x_{i,j}^{\text{Dis}} - x_j^D|}, & \text{if obstacle exists} \\ 1, & \text{otherwise.} \end{cases} \quad (2d)$$

Specifically, α and β in the distance sensing function (2a) are parameters that control the slope and the maximum effective range of the sensor, respectively. Specifically, α determines the steepness of the sigmoid function, with larger values resulting in a sharper transition from high to low-sensing probability as the distance increases. On the other hand, β sets the threshold distance beyond which the sensing probability rapidly decreases, effectively defining the maximum range of the sensor. In the pan sensing function (2b), ι and η determine the slope and the maximum angle of detection, respectively, considering the horizontal deviation relative to the sensor's central axis. Similarly, δ and ϱ in the tilt sensing function (2c) control the slope and the vertical deviation at which the sensor has 50% of its maximum coverage. Lastly, G in the obstacle sensing function (2d) is a variable that depends on the type of obstacle and affects the sensing probability in the presence of obstacles. A 3-D visualization of probabilistic coverage model is presented in Fig. 1.

The global sensing coverage of the RSNs for a target sensing area A_i^T is calculated as

$$GC(A_i^T, \mathcal{S}) = 1 - \prod_{j=1}^{|\mathcal{S}|} [1 - C(A_i^T, s_j)] \quad (3)$$

which represents the probability of covering A_i^T by the sensors deployed in $\mathcal{S}$.

D. Problem Formulation

The objective of the proposed system is to optimize the sensing performance for CAVs by efficiently utilizing the multilevel and multisource sensing information provided by the RSNs. The optimization problem aims to maximize the sensing accuracy and sensing range while minimizing the resource utilization, subject to various constraints.

Let $\mathcal{A}$ denote the set of target sensing areas, and $\mathcal{S}$ represent the set of deployed sensors in the RSNs. The decision variables are defined as follows.

- 1) $x_{c,v,m} \in \{0, 1\}$: Binary variable indicating whether sensing content c is transmitted from RSU m to CAV v .

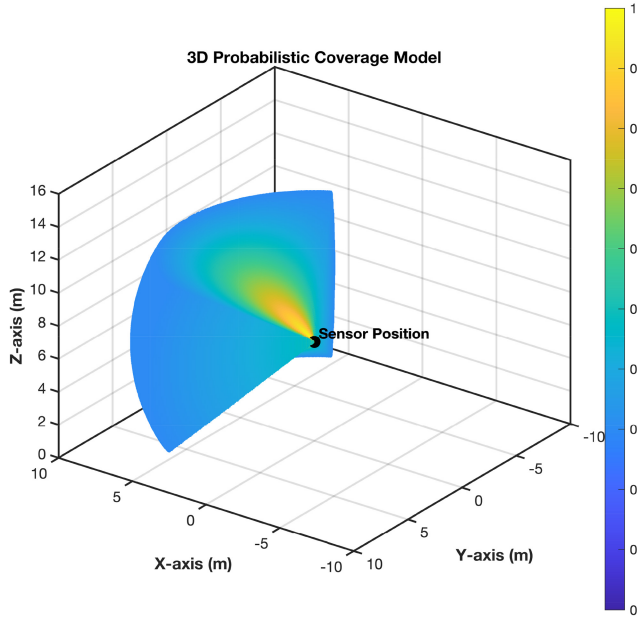


Fig. 1. 3-D visualization of the probabilistic coverage model for RSNs.

- 2) $y_{c,m}^k \in \{0, 1\}$: Binary variable indicating whether sensing content c is processed using resource type k on RSU m .
- 3) $z_{c,v}^k \in \{0, 1\}$: Binary variable indicating whether sensing content c is processed using resource type k on CAV v .
- 4) $n_i^\gamma \in \{0, 1, \dots, N_{i,max}^D\}$: Integer variable representing the number of sensors deployed at location p_i^D .

The multiobjective optimization problem can be formulated as follows:

$$\max \sum_{A_i^T \in \mathcal{A}} GC(A_i^T, \mathcal{S}) \cdot w_i \quad (4a)$$

$$\max \sum_{c \in \mathcal{C}} \sum_{v \in \mathcal{V}} \sum_{m \in \mathcal{M}} x_{c,v,m} \cdot q_c \quad (4b)$$

$$\text{s.t.} \quad \sum_{c \in \mathcal{C}} x_{c,v,m} \cdot s_c \leq b_{v,m} \quad \forall v \in \mathcal{V}, m \in \mathcal{M} \quad (4c)$$

$$\sum_{c \in \mathcal{C}} y_{c,m}^k \cdot p_c^k \leq r_m^k \quad \forall m \in \mathcal{M}, k \in \{1, \dots, K\} \quad (4d)$$

$$\sum_{c \in \mathcal{C}} z_{c,v}^k \cdot p_c^k \leq r_v^k \quad \forall v \in \mathcal{V}, k \in \{1, \dots, K\} \quad (4e)$$

$$\sum_{c \in \mathcal{C}} x_{c,v,m} = \sum_{k=1}^K (y_{c,m}^k + z_{c,v}^k) \quad \forall v \in \mathcal{V}, m \in \mathcal{M} \quad (4f)$$

$$\sum_{i=1}^M n_i^\gamma \leq N_{\max} \quad (4g)$$

$$x_{c,v,m}, y_{c,m}^k, z_{c,v}^k \in \{0, 1\} \quad \forall c \in \mathcal{C}, v \in \mathcal{V}, m \in \mathcal{M} \quad (4h)$$

$$n_i^\gamma \in \{0, 1, \dots, N_{i,max}^D\} \quad \forall i \in \{1, \dots, M\}. \quad (4i)$$

Equation (4a) maximizes the weighted sum of the global sensing coverage $GC(A_i^T, \mathcal{S})$ for each target sensing area A_i^T , where w_i represents the importance weight of area A_i^T . Equation (4b) maximizes the total amount of sensing content

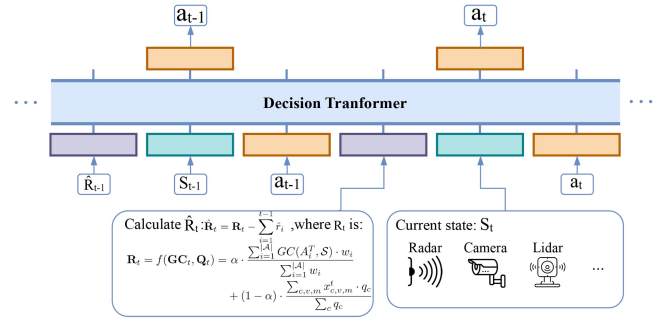


Fig. 2. Architecture of the DT model.

transmitted, where q_c represents the quality or importance of sensing content c .

Constraint (4c) ensures that the total amount of sensing content transmitted from RSU m to CAV v does not exceed the available bandwidth $b_{v,m}$. Equations (4d) and (4e) guarantee that the processing of sensing content on RSUs and CAVs does not exceed the available computing resources r_m^k and r_v^k , respectively. Equation (4f) ensures that the transmitted sensing content is processed either on the RSU or the CAV. Equation (4g) ensures that the total number of deployed sensors does not exceed the maximum allowed number $N_{\max}$. Equation (4h) defines the binary nature of the decision variables. Finally, (4i) defines the range of the sensor deployment variable n_i^γ .

IV. DECISION TRANSFORMER-BASED COOPERATIVE SENSING STRATEGY

This section introduces the proposed approach for optimizing cooperative sensing using DT. It explains how the decision-making process is formulated as a sequence and details the architecture of the DT model. The training process and policy optimization using the transformer model are also described.

A. Modeling Decision Making as a Sequence Problem

The proposed DT-based approach relies on the construction of decision-making sequences to optimize the cooperative sensing strategy [15]. These sequences are formulated as a combination of states $\mathbf{s}_t$, actions $\mathbf{a}_t$, and cumulative rewards $\mathbf{R}_t$. Fig. 2 illustrates the architecture of DT where states, actions, and return are fed into linear embeddings. The state $\mathbf{s}_t$ encapsulates the relevant information about the environment, including the set of deployed sensors $\mathcal{S}$, target sensing areas $\mathcal{A}$, available sensing content $\mathcal{C}$, and sensing probabilities $\mathbf{P}_t$. Formally, the state can be represented as $\mathbf{s}_t = \{\mathcal{S}, \mathcal{A}, \mathcal{C}, \mathbf{P}_t\}$ where $\mathbf{P}_t = [P_{i,j}^t]$ is a matrix of sensing probabilities, with $P_{i,j}^t$ denoting the sensing probability of sensor s_j for target area A_i^T at time t .

The action $\mathbf{a}_t$ represents the decision variables for sensor selection, content transmission, and resource allocation, as defined as

$$\mathcal{A}_t = \{\mathbf{x}_t, \mathbf{y}_t, \mathbf{z}_t, \mathbf{n}_t\} \quad (5)$$

where $\mathbf{x}_t = [x_{c,v,m}^t] \in \{0, 1\}^{|\mathcal{C}| \times |\mathcal{V}| \times |\mathcal{M}|}$, $\mathbf{y}_t = [y_{c,m}^{k,t}] \in \{0, 1\}^{|\mathcal{C}| \times |\mathcal{M}| \times K}$, $\mathbf{z}_t = [z_{c,v}^{k,t}] \in \{0, 1\}^{|\mathcal{C}| \times |\mathcal{V}| \times K}$, and $\mathbf{n}_t = [n_i^{\gamma,t}] \in \{0, 1, \dots, N_{i,\max}^D\}^M$ are the decision variables at time t .

The action space $\mathcal{A}_t$ includes binary decision variables $\mathbf{x}_t$, $\mathbf{y}_t$, and $\mathbf{z}_t$ for sensing content transmission and processing, and integer variable $\mathbf{n}_t$ for sensor deployment.

The cumulative reward $\mathbf{R}_t$ is a weighted combination of the global sensing coverage $\mathbf{GC}_t$ and the transmitted sensing content $\mathbf{Q}_t$. We define the reward function $f(\cdot)$ as follows:

$$\mathbf{R}_t = f(\mathbf{GC}_t, \mathbf{Q}_t) = \alpha \cdot \frac{\sum_{i=1}^{|\mathcal{A}|} GC(A_i^T, S) \cdot w_i}{\sum_{i=1}^{|\mathcal{A}|} w_i} + (1 - \alpha) \cdot \frac{\sum_{c,v,m} x_{c,v,m}^t \cdot q_c}{\sum_c q_c} \quad (6)$$

where $([\sum_{i=1}^{|\mathcal{A}|} GC(A_i^T, S) \cdot w_i] / [\sum_{i=1}^{|\mathcal{A}|} w_i])$ represents the weighted average of the global sensing coverage, considering the importance weights w_i for each target sensing area A_i^T . $([\sum_{c,v,m} x_{c,v,m}^t \cdot q_c] / [\sum_c q_c])$ represents the normalized sum of the transmitted sensing content, considering the quality or importance q_c of each sensing content c . $\alpha \in [0, 1]$ is a tunable parameter that controls the tradeoff between the global sensing coverage and the transmitted sensing content. A higher value of α will place more emphasis on maximizing the global sensing coverage, while a lower value will prioritize the transmitted sensing content.

B. Learning Optimal Sensing Strategies With DT

The DT model is trained on a diverse set of simulated scenarios, each representing different environmental conditions, traffic patterns, and cooperative sensing challenges [28]. The training process is critical for the model to learn the intricacies of effective sensor selection and content transmission [17], and to understand how different actions impact the global sensing coverage and the quality of transmitted sensing content.

During training, the DT model learns to optimize the cumulative reward $\mathbf{R}_t$ by minimizing the following loss function as shown as:

$$\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_{\theta}(\mathbf{a}_t | \mathbf{s}_{\leq t}, \mathbf{R}_{\geq t}) \quad (7)$$

where θ represents the model parameters, T is the sequence length, and $p_{\theta}(\mathbf{a}_t | \mathbf{s}_{\leq t}, \mathbf{R}_{\geq t})$ is the probability of taking action $\mathbf{a}_t$ given the state sequence $\mathbf{s}_{\leq t}$ and the reward sequence $\mathbf{R}_{\geq t}$.

The DT model utilizes a multihead attention mechanism, which allows it to focus on different aspects of the input sequence and capture various facets of the cooperative sensing problem. The multihead attention mechanism is formulated as

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (8a)$$

$$\text{where head}_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right) \quad (8b)$$

where Q , K , and V represent the query, key, and value matrices, respectively, and W_i^Q , W_i^K , and W_i^V are learned

projection matrices for each head. The attention mechanism within each head is computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (9)$$

where d_k is the dimension of the key vectors, acting as a scaling factor.

By leveraging the multihead attention mechanism, the DT model can simultaneously consider multiple aspects of the cooperative sensing problem, such as the sensing capabilities of the sensors, the quality of the sensing content, and the available resources on the RSUs and CAVs.

C. Optimizing Cooperative Sensing Strategies Using Transformers

The DT model optimizes the cooperative sensing policy by analyzing historical sequences of states, actions, and rewards. The model learns to predict the optimal actions that maximize the expected cumulative reward, considering the current state and the learned patterns from previous experiences [32]. The policy optimization is formulated as

$$\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) = \arg \max_{\mathbf{a}_t} \mathbb{E}[\mathbf{R}_t | \mathbf{s}_t, \mathbf{a}_t, \theta] \quad (10)$$

where π_{θ} is the policy parameterized by the DT model parameters θ , $\mathbf{s}_t$ is the current state, $\mathbf{a}_t$ is the action taken at time t , and $\mathbf{R}_t$ is the expected cumulative reward.

The optimization process is guided by the multiobjective functions defined in the problem formulation, aiming to maximize the global sensing coverage and the quality of transmitted sensing content. The DT model's performance is continuously evaluated against these objectives to ensure that the generated cooperative sensing strategies are effective and applicable in real-world scenarios.

V. IMPROVED DT THROUGH REWARD REDISTRIBUTION

The section presents a novel technique for enhancing the performance of the DT-based cooperative sensing strategy. It introduces a reshaped reward function and a bi-level optimization framework to effectively handle long-delayed rewards. The advanced gradient approximation method used to efficiently solve the optimization problem is also detailed.

To further clarify the improved DT algorithm, we present a flowchart of the process in Fig. 3. The flowchart illustrates the initialization of the policy model and shaping reward function parameters, the iterative updating of these parameters using the advanced gradient approximation technique, and the bi-level optimization process that involves updating θ based on the validation loss and ϕ based on the training loss. The algorithm continues until the convergence criteria are met, resulting in an optimized policy model and shaping reward function for cooperative sensing in resource-constrained environments.

A. Enhancing DT With Reshaped Rewards

In the DT-based cooperative sensing strategy for resource-constrained environments, the effective redistribution of

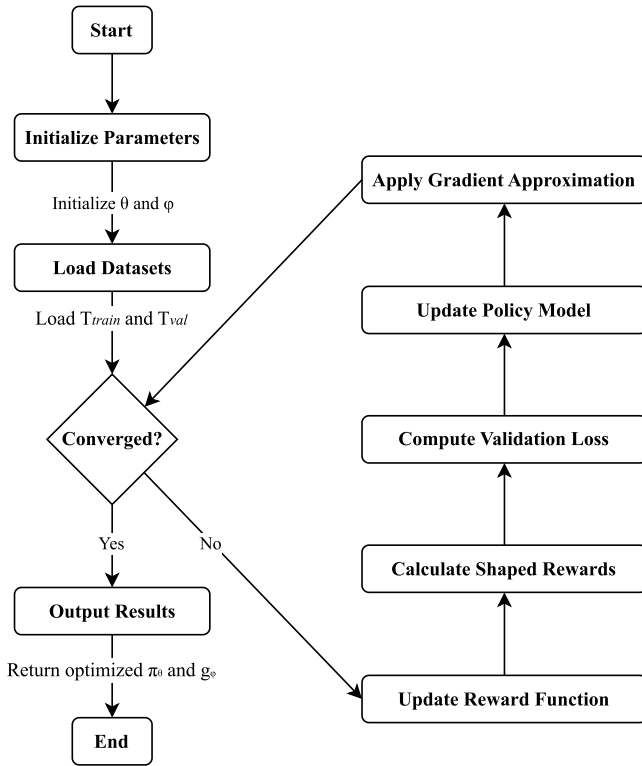


Fig. 3. Flowchart of the improved DT algorithm for cooperative sensing in resource-constrained environments.

rewards along the decision-making trajectories is crucial. Given a set of offline decision trajectories

$$\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_N\} \quad (11)$$

where each trajectory τ_i represents a sequence of states, actions, and rewards, the objective is to redistribute the cumulative impact of the trajectory at each timestep to facilitate learning an optimal cooperative sensing strategy.

We introduce a shaping reward function g_ϕ , which is essential for recalibrating the rewards at each timestep in the decision-making trajectory. The reshaped reward $\hat{r}_t$ at timestep t is formulated as

$$\hat{r}_t = \hat{R}_{\text{step}}(\mathbf{s}_t, \mathbf{a}_t) = g_\phi(\mathbf{s}_t, \mathbf{a}_t) \quad (12)$$

where $\mathbf{s}_t, \mathbf{a}_t \in \tau$, and $g_\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the shaping reward function parameterized by ϕ . This function assigns a revised reward value to each state-action pair within a trajectory.

Since the DT model utilizes the cumulative reward-to-go $\mathbf{R}_t$ instead of individual rewards r_t , we calculate the shaped cumulative rewards based on these revised rewards. The shaped cumulative reward $\hat{\mathbf{R}}_t$ is modeled as

$$\hat{\mathbf{R}}_t = \mathbf{R}_t - \sum_{i=1}^{t-1} \hat{r}_i \quad (13)$$

which ensures the consistency of the total reward across the trajectory before and after redistribution. The shaping reward function g_ϕ is designed to capture the intricate characteristics of the environment, such as the sensing capabilities of the

sensors, the quality of the sensing content, and the available resources on the RSUs and CAVs.

The key idea behind the reshaped reward function is to redistribute the cumulative rewards along the decision sequence in a way that provides more informative feedback to the agent at each step. By assigning higher rewards to actions that lead to better long-term outcomes and lower rewards to those that result in suboptimal future states, the reshaped reward function helps the agent learn more efficiently and make better decisions over time. For example, consider a scenario where the agent has to decide between two sensing actions: one that provides high-immediate rewards but leads to poor future performance and another that yields lower immediate rewards but enables better long-term outcomes. The reshaped reward function would assign higher rewards to the latter action, encouraging the agent to prioritize long-term benefits over short-term gains. This intuitive reshaping of rewards facilitates the effective propagation of feedback along the decision sequence, guiding the agent toward optimal sensing strategies in resource-constrained environments.

B. Bilevel Optimization Framework

Incorporating a bi-level optimization framework in the DT-based cooperative sensing strategy involves two interconnected tasks: optimizing the policy model, content transmission, and resource allocation, and refining the shaping reward function to guide this optimization [5]. To determine the optimal shaping reward function for the DT-based cooperative sensing strategy, we introduce a comprehensive end-to-end, bi-directional bi-level optimization framework. This framework enables the trained reward function to guide the training of the policy model and allows the policy model to iteratively refine the reward function, creating a dynamic loop of bi-directional feedback and joint optimization.

The core concept is to use a validation set to evaluate the effect of redistributed rewards on the policy model's performance, guiding the reward function's optimization accordingly [4]. Consequently, all trajectory data $\mathcal{T}$ are categorized into two sets: 1) the training set $\mathcal{T}_{\text{train}}$ for policy model optimization and 2) the validation set $\mathcal{T}_{\text{val}}$ for reward function optimization.

1) *Lower Level Optimization*: The lower level optimization focuses on enhancing the policy model π_θ , which learns optimal cooperative sensing strategies from a given set of decision-making trajectories $\mathcal{T}_{\text{train}}$. This process involves training the policy model to adapt to the reshaped rewards, ensuring it can accurately reflect the objectives and constraints of the cooperative sensing problem. The policy model π_θ is trained to make decisions that align with the reshaped reward function.

2) *Upper Level Optimization*: The upper level task focuses on optimizing the shaping reward function g_ϕ [9]. Its aim is to discover a reward function configuration that significantly enhances the performance of the policy model on a separate validation set $\mathcal{T}_{\text{val}}$, thereby validating the effectiveness of the reshaped rewards [21]. The shaping reward function g_ϕ is adjusted to maximize the policy model's decision-making performance.

3) *Bi-Level Optimization Problem Formulation*: We define the bi-level optimization problem as minimizing the validation loss $\mathcal{L}_{\text{val}}(\theta^*(\phi), \phi)$, subject to the constraint that the total reward remains consistent before and after the reward redistribution as shown as

$$\min_{\phi} \quad \mathcal{L}_{\text{val}}(\theta^*(\phi), \phi) \quad (14a)$$

$$\text{s.t.} \quad \mathbf{R}_{\tau} = \sum_{t=1}^{|\tau|} \hat{r}_t \quad \forall \tau \in \mathcal{T}_{\text{val}} \quad (14b)$$

$$\theta^*(\phi) = \arg \min_{\theta} \mathcal{L}_{\text{train}}(\theta, \phi) \quad (14c)$$

where ϕ is the upper level variable (shaping reward function parameters), θ is the lower level variable (policy model parameters), $\mathbf{R}_{\tau}$ is the cumulative reward of trajectory τ , and $\hat{r}_t = g_{\phi}(\mathbf{s}_t, \mathbf{a}_t)$ is the shaped reward at each timestep.

To transform this constrained bi-level optimization problem into an unconstrained one, we introduce a quadratic penalty function. The objective function can then be rewritten as

$$\min_{\phi} \quad \mathcal{L}_{\text{val}}(\theta^*(\phi), \phi) + \lambda \sum_{\tau \in \mathcal{T}_{\text{val}}} \left(\mathbf{R}_{\tau} - \sum_{t=1}^{|\tau|} \hat{r}_t \right)^2 \quad (15a)$$

$$\text{s.t.} \quad \theta^*(\phi) = \arg \min_{\theta} \mathcal{L}_{\text{train}}(\theta, \phi) \quad (15b)$$

where λ is a hyperparameter balancing the original objective function and the penalty term. This bi-level optimization framework enables the DT-based cooperative sensing strategy to learn effective reward redistribution and optimize the policy model for improved decision making in resource-constrained environments.

C. Gradient Approximation for DT Optimization

In the framework of the DT-based cooperative sensing strategy for resource-constrained environments, the bi-level optimization process is crucial to the model's efficiency. However, accurately computing the gradient for the upper level shaping reward function g_{ϕ} can be challenging due to its tight coupling with the lower level policy optimization π_{θ} [21]. This complexity often necessitates multiple iterations to identify the optimal policy at the lower level, leading to a potentially time-consuming process. To address this, we adopt an alternating optimization technique, which efficiently separates the parameters θ and ϕ , thus expediting the optimization process.

We propose a straightforward approximation scheme for this purpose, as shown in the following equation:

$$\theta^*(\phi) \approx \theta'(\phi) = \theta - \alpha_{\theta} \nabla_{\theta} \mathcal{L}_{\text{train}}(\theta, \phi) \quad (16)$$

where θ represents the current policy as trained by the algorithm, and α_{θ} is the learning rate for a single step of lower level policy optimization. Drawing from meta-learning principles, the idea is to adapt θ using just a single training step, rather than fully solving the lower level optimization problem.

Algorithm 1 Improved DT for Cooperative Sensing

Require: Learning rates α_{θ} for the policy model and α_{ϕ} for the shaping reward function, training dataset $\mathcal{T}_{\text{train}}$, and validation dataset $\mathcal{T}_{\text{val}}$

- 1: Initialize the policy model parameters θ and shaping reward function parameters ϕ
- 2: **while** not converged **do**
- 3: Update ϕ by descending along its gradient

$$\phi \leftarrow \phi - \alpha_{\phi} \left(\nabla_{\phi} \mathcal{L}_{\text{val}}(\theta'(\phi), \phi) + \lambda \nabla_{\phi} \sum_{\tau \in \mathcal{T}_{\text{val}}} \left(\mathbf{R}_{\tau} - \sum_{t=1}^{|\tau|} g_{\phi}(\mathbf{s}_t, \mathbf{a}_t) \right)^2 \right).$$

- 4: Update θ by descending along its gradient

$$\theta \leftarrow \theta - \alpha_{\theta} \nabla_{\theta} \mathcal{L}_{\text{train}}(\theta, \phi)$$

5: **end while**

- 6: **return** the optimized policy model π_{θ} and the shaping reward function g_{ϕ}

With this scheme, the computation of the upper level gradient is effectively approximated as follows:

$$\begin{aligned} & \nabla_{\phi} \left[\mathcal{L}_{\text{val}}(\theta^*(\phi), \phi) + \lambda \sum_{\tau \in \mathcal{T}_{\text{val}}} \left(\mathbf{R}_{\tau} - \sum_{t=1}^{|\tau|} \hat{r}_t \right)^2 \right] \\ & \approx \nabla_{\phi} \mathcal{L}_{\text{val}}(\theta'(\phi), \phi) + \lambda \nabla_{\phi} \sum_{\tau \in \mathcal{T}_{\text{val}}} \left(\mathbf{R}_{\tau} - \sum_{t=1}^{|\tau|} g_{\phi}(\mathbf{s}_t, \mathbf{a}_t) \right)^2. \end{aligned} \quad (17)$$

The first term of the approximated upper gradient can be expanded using the chain rule as shown as

$$\begin{aligned} \nabla_{\phi} \mathcal{L}_{\text{val}}(\theta'(\phi), \phi) &= -\alpha_{\theta} \nabla_{\phi, \theta}^2 \mathcal{L}_{\text{train}}(\theta, \phi) \nabla_{\theta'} \mathcal{L}_{\text{val}}(\theta', \phi) \\ &\quad + \nabla_{\phi} \mathcal{L}_{\text{val}}(\theta', \phi) \end{aligned} \quad (18)$$

where $\theta' = \theta - \alpha_{\theta} \nabla_{\theta} \mathcal{L}_{\text{train}}(\theta, \phi)$ represents the parameters for a one-step forward policy model.

To simplify the computation, we apply Taylor's formula for approximating the product of first and second-order gradients. Let η be a small scalar

$$\begin{aligned} & \nabla_{\phi, \theta}^2 \mathcal{L}_{\text{train}}(\theta, \phi) \nabla_{\theta'} \mathcal{L}_{\text{val}}(\theta', \phi) \\ & \approx \frac{1}{2\eta} \left(\nabla_{\phi} \mathcal{L}_{\text{train}}(\theta + \eta \nabla_{\theta'} \mathcal{L}_{\text{val}}(\theta', \phi), \phi) \right. \\ & \quad \left. - \nabla_{\phi} \mathcal{L}_{\text{train}}(\theta - \eta \nabla_{\theta'} \mathcal{L}_{\text{val}}(\theta', \phi), \phi) \right). \end{aligned} \quad (19)$$

This approximation scheme ensures efficient computation and accelerates the overall optimization process in the DT framework for cooperative sensing, crucial for handling complex scenarios with delayed rewards in resource-constrained environments.

The proposed improved DT approach for cooperative sensing is summarized in Algorithm 1. The algorithm iteratively updates the policy model parameters θ and the shaping

reward function parameters ϕ using the advanced gradient approximation technique. It considers the shaped rewards and the consistency constraint while minimizing the validation and training losses. The optimized policy model π_θ and the shaping reward function g_ϕ are returned, enabling efficient and effective learning of cooperative sensing strategies in resource-constrained environments.

VI. SIMULATION RESULTS AND DISCUSSION

In this section, this article presents a comprehensive evaluation of the proposed improved DT approach through extensive simulations. The simulation setup and parameter settings are described, followed by a detailed analysis of the performance results.

A. Parameter Setting

To evaluate the performance of the proposed improved DT approach, we conduct extensive simulations using a custom-built simulator developed in Python Anaconda 3.8.8. The simulator models the deployment and performance of RSNs in realistic urban environments, utilizing the OpenAI Gym [33] framework for defining the simulation environment and agent interactions. Traffic flow patterns are modeled using historical taxi data [34] from Roma to ensure the simulated traffic closely resembles real-world conditions. The simulated environment integrates these data sources to create a detailed 3-D model of the road network, including buildings, vegetation, and other static obstacles.

The experiments are executed 100 times on a single core of an Intel i7 CPU operating at 2.8 GHz. We consider three key metrics to assess the effectiveness and efficiency of our proposal: 1) average detection accuracy [8]; 2) global sensing coverage [8]; and 3) average data size [35]. The proposed improved DT approach is compared against established RL methods, including rainbow DRL, soft actor-critic RL, and proximal policy optimization, which have demonstrated state-of-the-art performance in various application areas.

Benchmarks: We compare our improved DT approach with established RL methods, including Rainbow DRL [36], soft actor-critic RL [37], and proximal policy optimization [38]. These benchmarks have demonstrated state-of-the-art performance across various application domains, making them suitable for comparison. It is important to note that while these RL methods use the reward function for both training and performance assessment, our improved DT algorithm employs the reward function solely for evaluating performance, facilitating a meaningful comparison.

In the simulations, we evaluate the performance of the proposed system under diverse conditions. The parameter settings for the experiment are set with [32], [39], and [40]. The simulation parameters used in our experiments are summarized in Table II. Usually, when traffic density ranges from 0 to 12 500 vehicles per hour, it can typically be categorized into three levels: 1) normal (0–4000 veh/hr); 2) moderate (4001–8000 veh/hr); and 3) crowded (8001–12 500 veh/hr). For example, the value of α was determined through a combination of domain knowledge and empirical

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Simulation area	5 km × 5 km
Number of CAVs	50–200
Number of RSUs	10–50
CAV sensing range	100 m
CAV velocity	30–60 km/h
Available bandwidth	10–50 Mbps
Computation resource per RSU	10–50 GHz
Computation resource per CAV	1–5 GHz
Importance weight α	0.5
Learning rate α_θ	0.001
Learning rate α_ϕ	0.01

analysis. We considered the relative importance of sensing coverage and content quality in the context of CAVs and conducted experiments with different values of α to identify the one that struck the best balance for our specific problem setting. In our simulations, we found that setting α to 0.5 provided the optimal tradeoff. These parameters have been carefully selected to reflect realistic scenarios and to ensure a comprehensive evaluation of the proposed improved DT approach for cooperative sensing in resource-constrained environments.

B. Performance Analyses

Fig. 4(a) and (b) illustrate the impact of the number of available sensors and traffic density on the average sensing accuracy of our proposed improved DT approach compared to three state-of-the-art RL methods. As the number of sensors increases, the sensing accuracy improves for all methods, with our proposal consistently outperforming the others, achieving 96.56% accuracy with 25 sensors. Conversely, increasing traffic density leads to a decline in sensing accuracy due to the increased complexity and dynamic nature of the environment. Nonetheless, our approach maintains higher accuracy compared to the other methods, even in high-density scenarios, demonstrating its robustness and adaptability. These superior results can be attributed to the enhanced reward redistribution and bi-level optimization techniques employed in our improved DT approach.

Fig. 5 presents the impact of the number of available sensors and traffic density on the sensing coverage achieved by our proposed improved DT approach and three state-of-the-art RL methods. As depicted in Fig. 5(a), increasing the number of sensors leads to improved coverage for all methods. Our proposal consistently achieves higher coverage compared to the other methods, reaching 78.56% with 25 sensors, while Rainbow DRL, Soft AC RL, and Policy Optimization achieve 73.35%, 66.83%, and 64.35%, respectively. Fig. 5(b) shows the impact of traffic density on sensing coverage. Interestingly, our approach maintains a stable and slightly increasing coverage trend as traffic density increases, reaching 79% at 12 500 vehicles per hour. In contrast, the other methods exhibit a decline in coverage at high-traffic densities, with Rainbow DRL, Soft AC RL, and Policy Optimization achieving 74.44%, 68.44%, and 65.44%, respectively. This demonstrates the

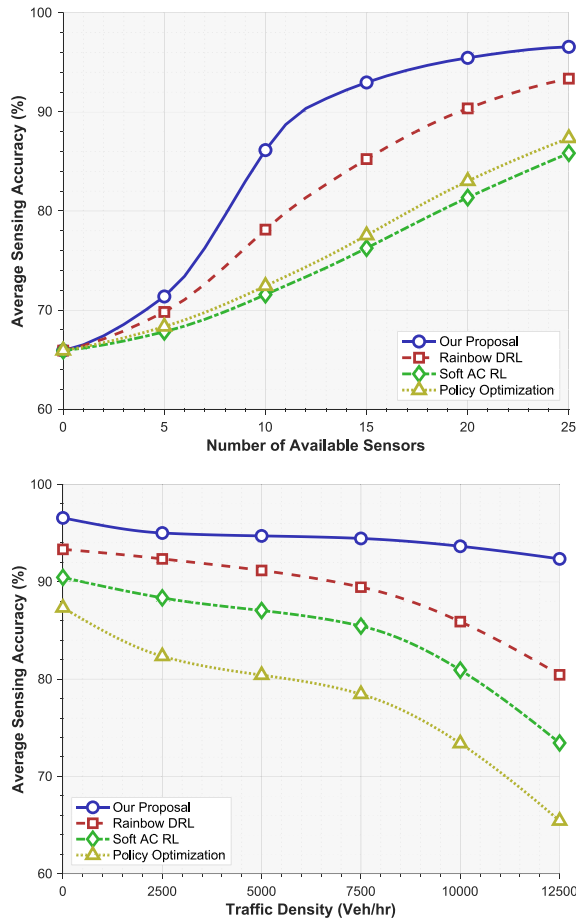


Fig. 4. Impact of the number of sensors and traffic density on the average sensing accuracy of the proposed improved DT approach.

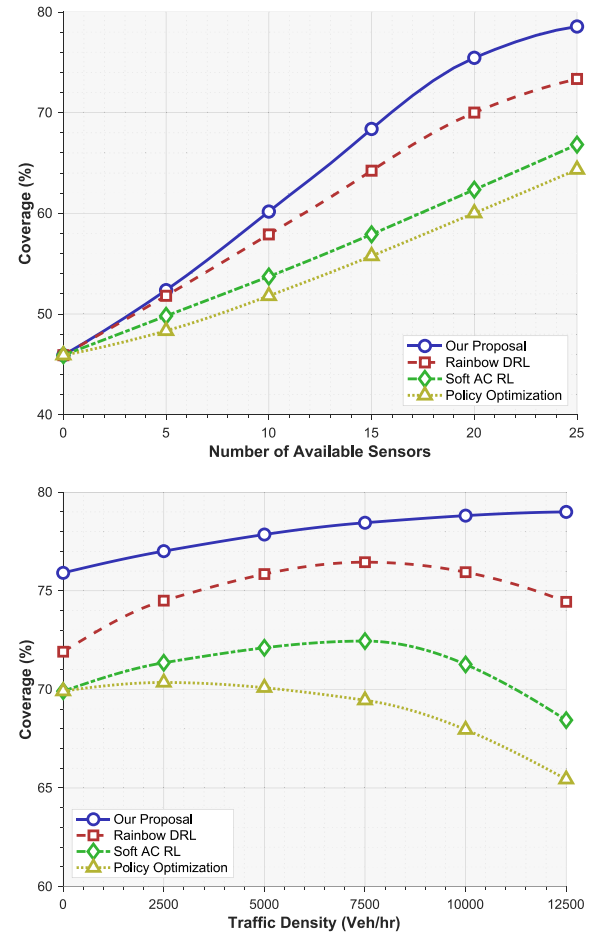


Fig. 5. Impact of the number of sensors and traffic density on the sensing coverage achieved by the proposed improved DT approach.

superior ability of our improved DT approach to adapt and maintain high-sensing coverage even in dense traffic scenarios, highlighting its potential for real-world deployment in intelligent transportation systems.

The impact of visibility on the performance of our proposed improved DT approach and the state-of-the-art RL methods is illustrated in Fig. 6. As visibility increases from 0.25 to 1, all methods demonstrate improved average sensing accuracy and coverage. However, our proposal consistently outperforms the other approaches across all visibility levels. At the lowest visibility of 0.25, our method achieves an accuracy of 75% and coverage of 65%, while the best performing alternative, Rainbow DRL, only reaches 70% accuracy and 60% coverage. The performance gap widens as visibility improves, with our approach attaining 96% accuracy and 82% coverage at full visibility, compared to 90% accuracy and 75% coverage for Rainbow DRL. This trend highlights the robustness of our improved DT method, which effectively leverages the available sensing information to maintain high performance even in challenging low-visibility scenarios. The superior performance can be attributed to the enhanced reward redistribution and bi-level optimization techniques, which enable the model to adapt and learn optimal sensing strategies under various visibility conditions. Besides, it is important to note that the visibility experiment not only evaluates our method's performance

in challenging perceptual conditions but also serves as a proxy for scenarios involving sensor failures. In low-visibility environments, some sensors may fail to function properly or provide accurate data, similar to the effects of sensor failures in real-world deployments. The superior performance of our improved DT approach in these conditions, as demonstrated by the higher sensing accuracy and coverage compared to the baselines, highlights its robustness and adaptability to sensor failures and noisy data.

Figs. 7 and 8 illustrate the relationship between traffic patterns, characterized by the percentages of safety-priority and efficiency-priority users, and the performance of our proposed improved DT approach in terms of average detection accuracy and coverage, respectively. As the proportion of safety-priority users increases, the detection accuracy improves, reaching its highest value of around 95% when all users prioritize safety. Conversely, a higher percentage of efficiency-priority users leads to lower detection accuracy, with the lowest value of approximately 75% observed when all users prioritize efficiency. The coverage exhibits a different trend, with the highest values of around 90% achieved when the majority of users prioritize efficiency, and the lowest coverage of about 70% occurring when most users prioritize safety. These results highlight the tradeoff between accuracy and coverage

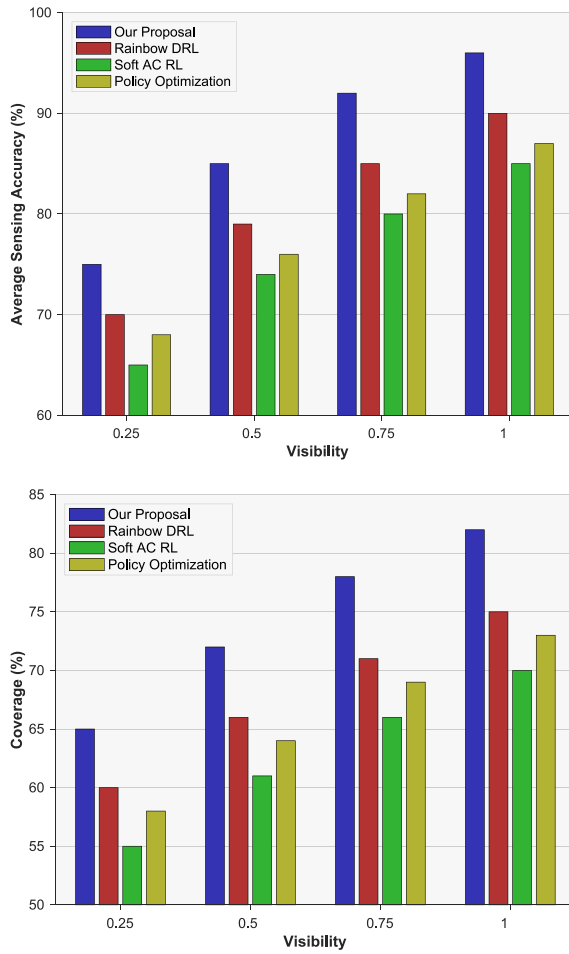


Fig. 6. Impact of visibility on the average sensing accuracy and coverage of the proposed improved DT approach.

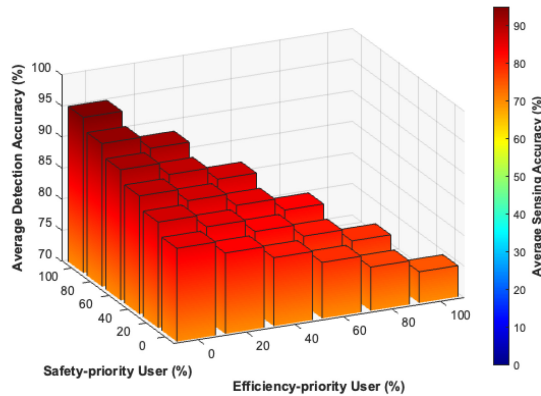


Fig. 7. Influence of different traffic patterns, characterized by the percentages of safety-priority and efficiency-priority users, on the average detection accuracy attained by the proposed improved DT approach.

in different traffic patterns, emphasizing the importance of adapting the cooperative sensing strategy to the specific needs of the users. Our improved DT approach demonstrates its ability to effectively balance these competing objectives and provide optimal performance based on the prevailing traffic pattern, underlining its potential for real-world deployment in diverse transportation scenarios.

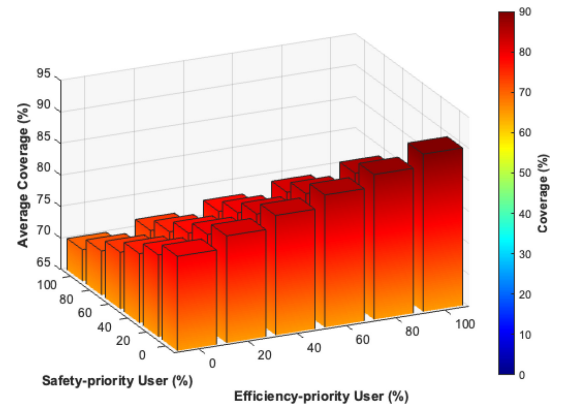


Fig. 8. Influence of different traffic patterns, characterized by the percentages of safety-priority and efficiency-priority users, on the sensing coverage attained by the proposed improved DT approach.

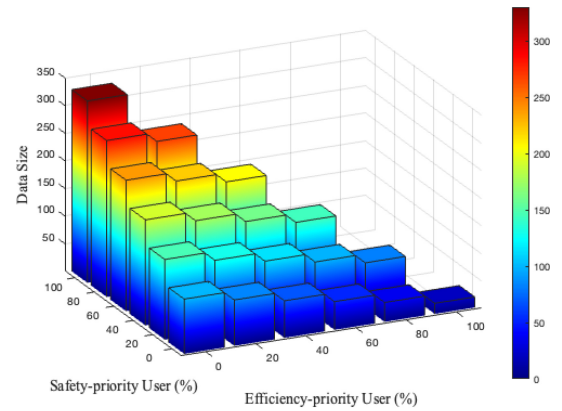


Fig. 9. Relationship between traffic patterns (percentages of safety-priority and efficiency-priority users) and the average data size transmitted in the proposed improved DT approach.

The relationship between traffic patterns and the average data size transmitted in our proposed improved DT approach is depicted in Fig. 9. As the percentage of safety-priority users increases, the average data size grows significantly, reaching a maximum of approximately 330 units when all users prioritize safety. This is due to the higher proportion of raw sensor data being transmitted to ensure the highest level of safety. In contrast, when the majority of users are efficiency-priority, the average data size decreases to around 20 units, as more processed and compressed result-level data is transmitted to optimize communication efficiency. The tradeoff between safety and efficiency is evident in the data size, with comfort-priority users falling in between, requiring feature-level data of moderate size. These findings underscore the adaptability of our improved DT approach in adjusting the transmitted data size based on the prevailing traffic pattern, ensuring optimal performance while balancing safety and efficiency requirements.

Fig. 10 compares the convergence trends of our proposed improved DT method and the standard DT approach without reward shaping. The cumulative reward achieved by both methods increases with the number of epochs, indicating the learning progress. However, our improved DT method consistently outperforms the standard DT approach, reaching

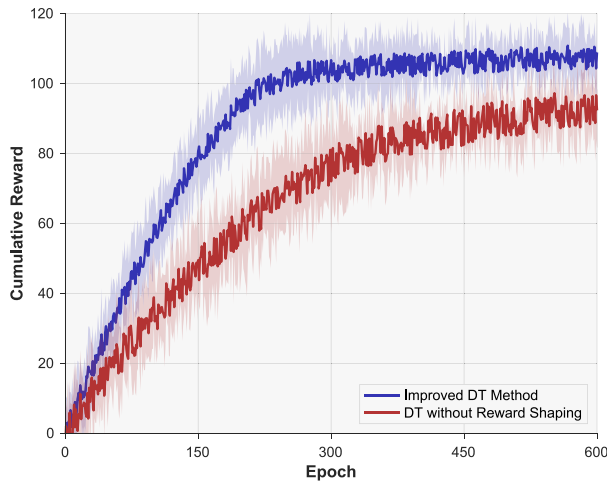


Fig. 10. Convergence trends and cumulative rewards of the proposed improved DT method compared to the standard DT approach without reward shaping, over multiple epochs.

a higher cumulative reward of around 108 units compared to 94 units achieved by the latter. Moreover, our method converges faster, requiring fewer epochs to attain its maximum performance. The shaded areas represent the variance in the cumulative reward across multiple runs, with our approach exhibiting lower variance and more stable learning. These results demonstrate the effectiveness of our reward shaping technique in enhancing the learning efficiency and performance of the DT model for cooperative sensing in resource-constrained environments.

To implement and adapt our improved DT framework in real-world CAV networks, several key considerations need to be addressed. First, our approach can be integrated into existing CAV architectures by mapping the inputs and outputs of our framework to the corresponding interfaces in the CAV system. Second, to handle the heterogeneous sensor environments in real-world deployments, the preprocessing and feature extraction stages of our framework can be extended to support multiple sensor modalities and data representations. Third, our approach can be expanded to leverage the computing and storage resources available in the infrastructure and cloud layers of the CAV ecosystem, enabling higher efficiency and scalability through task offloading and collaboration. While the modular and flexible design of our framework facilitates integration and adaptation to various CAV platforms and sensing configurations, further research and development efforts, including extensive field testing, standardization of interfaces and protocols, and collaboration with industry partners, are needed to fully realize its potential in large-scale real-world deployments.

VII. CONCLUSION

This article presented a novel cooperative sensing framework that leverages multisource and multilevel sensing information from RSNs to optimize the sensing performance of CAVs in resource-constrained environments. We have addressed the key challenges of integrating RSNs into the CAV ecosystem, including heterogeneous sensing requirements, inconsistencies in multisource sensor data, and efficient

resource utilization. Our proposed approach utilized an improved DT model that dynamically adapts to diverse driving conditions and efficiently fuses sensor data at various abstraction levels. To tackle the issue of long-delayed rewards, we introduced a reshaped reward function and a bi-level optimization framework that enables effective propagation of rewards along decision sequences. An advanced gradient approximation technique was employed to efficiently solve the optimization problem. Extensive simulations have demonstrated the superior performance of our improved DT approach compared to state-of-the-art RL methods in terms of sensing accuracy, coverage, and data efficiency under various traffic patterns and environmental conditions.

It is important to acknowledge the limitations of our proposed approach, particularly in terms of computational complexity and scalability. As the size of the RSN and the complexity of the driving conditions increase, the dimensionality of the state and action spaces in our DT model also grows, leading to longer training times and higher memory requirements. Future research directions include the incorporation of diverse sensing modalities, development of distributed cooperative sensing algorithms, and integration with advanced perception and decision-making techniques. Real-world demonstrations and validations will be crucial for bridging the gap between simulation and practical implementation. Exploring thermal cameras, ultrasonic sensors, and high-resolution radar could enhance the framework's robustness and adaptability. Besides, decentralized algorithms that enable efficient collaboration among vehicles and infrastructure, reducing reliance on central coordination, should be investigated. Integrating our approach with deep learning-based object detection, semantic segmentation, model predictive control, and game theory-based strategies could lead to more intelligent systems.

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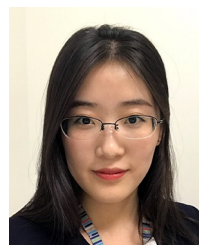
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